

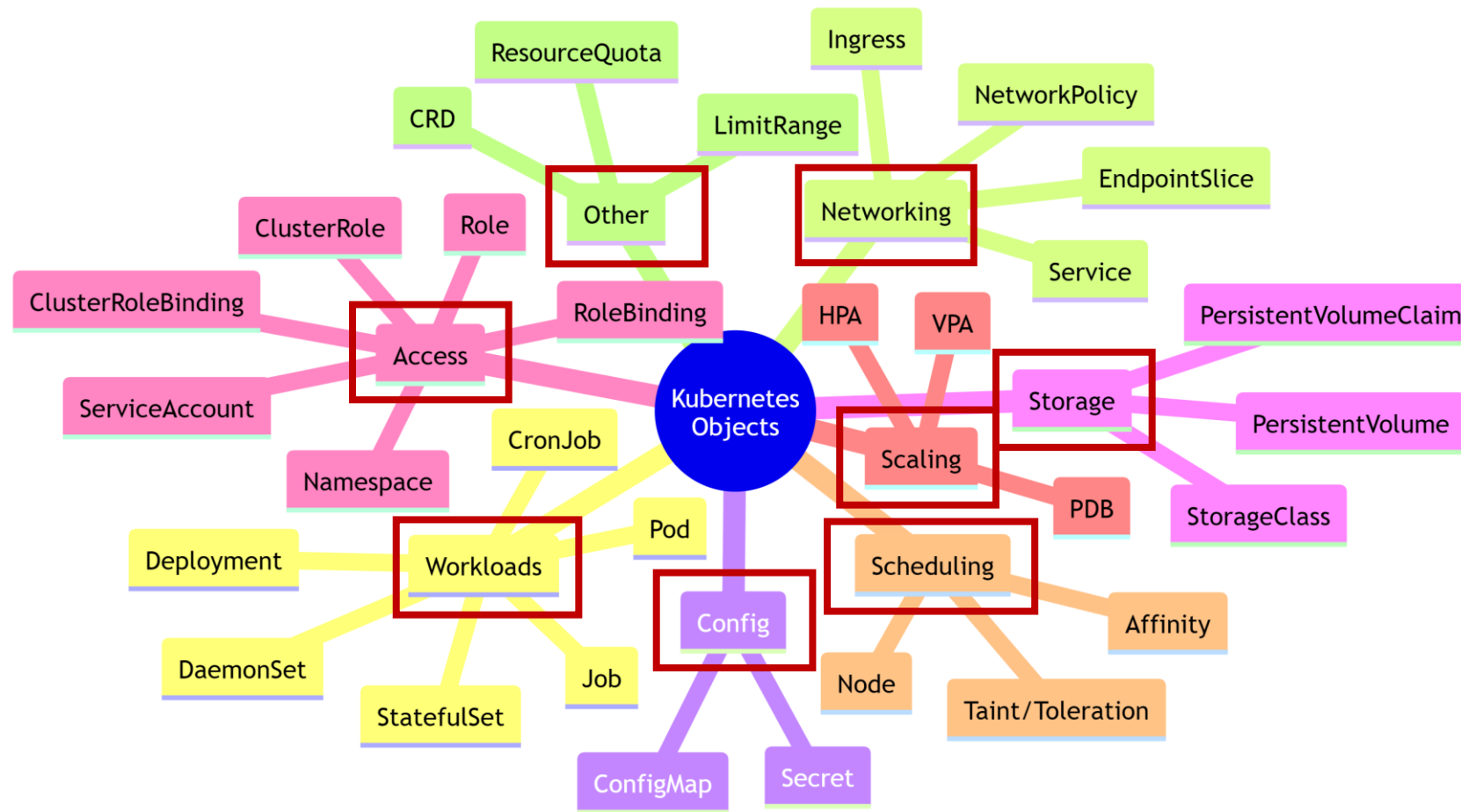
Implementing a (single node) cluster of Kafka by Kubernetes

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MLSECOPS IN ACTION COURSE

Learn Kubernetes Objects by Question

Learn Kubernetes Objects by Question



Questions?

Run a stateless web app	Deployment
Run a database or Kafka	StatefulSet
Run a log agent on every node	DaemonSet
Run a one-time migration	Job
Run a nightly backup	CronJob
Expose an app inside the cluster	Service (ClusterIP)
Expose an app outside the cluster	Service (NodePort / LoadBalancer)
Route HTTP by domain/path	Ingress
Give a pod a stable name	StatefulSet + Headless Service
Store a password	Secret
Store a config file	ConfigMap
Persist data across restarts	PVC (PersistentVolumeClaims)
Auto-scale on CPU	HPA (Horizontal Pod Autoscaling)
Define my own object type	CRD (custom resource definition)

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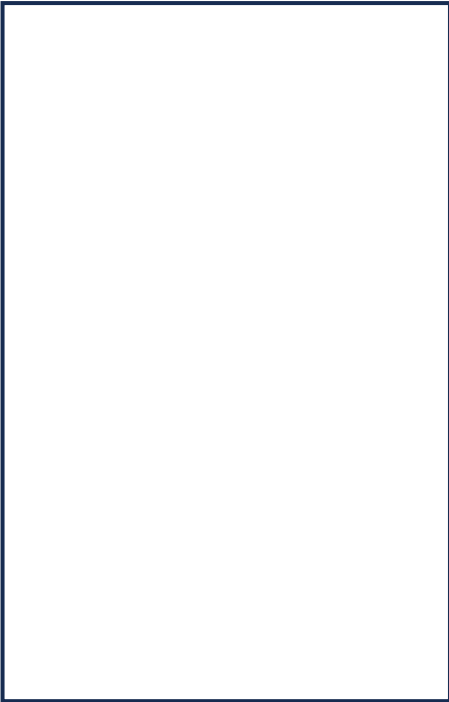
Our Task Architecture

Kubernetes Cluster

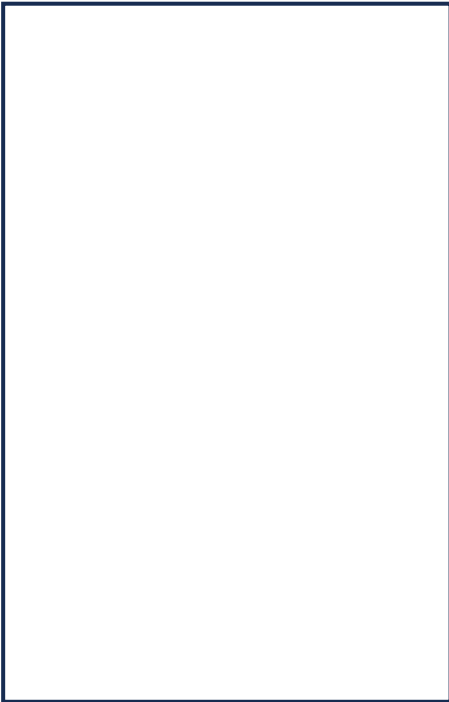


Control-Plane

K8S-Master



K8S-Worker-01



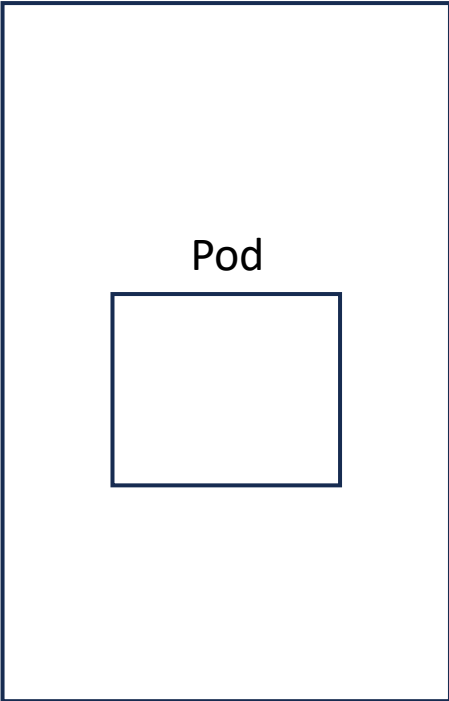
K8S-Worker-02

Kubernetes Cluster



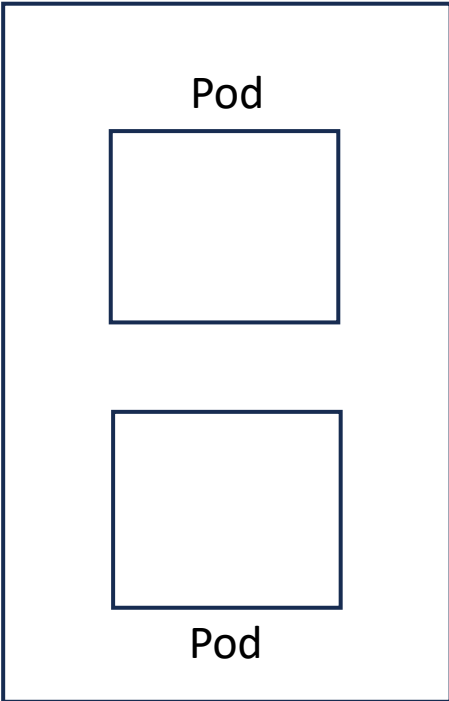
Control-Plane

K8S-Master



Pod

K8S-Worker-01

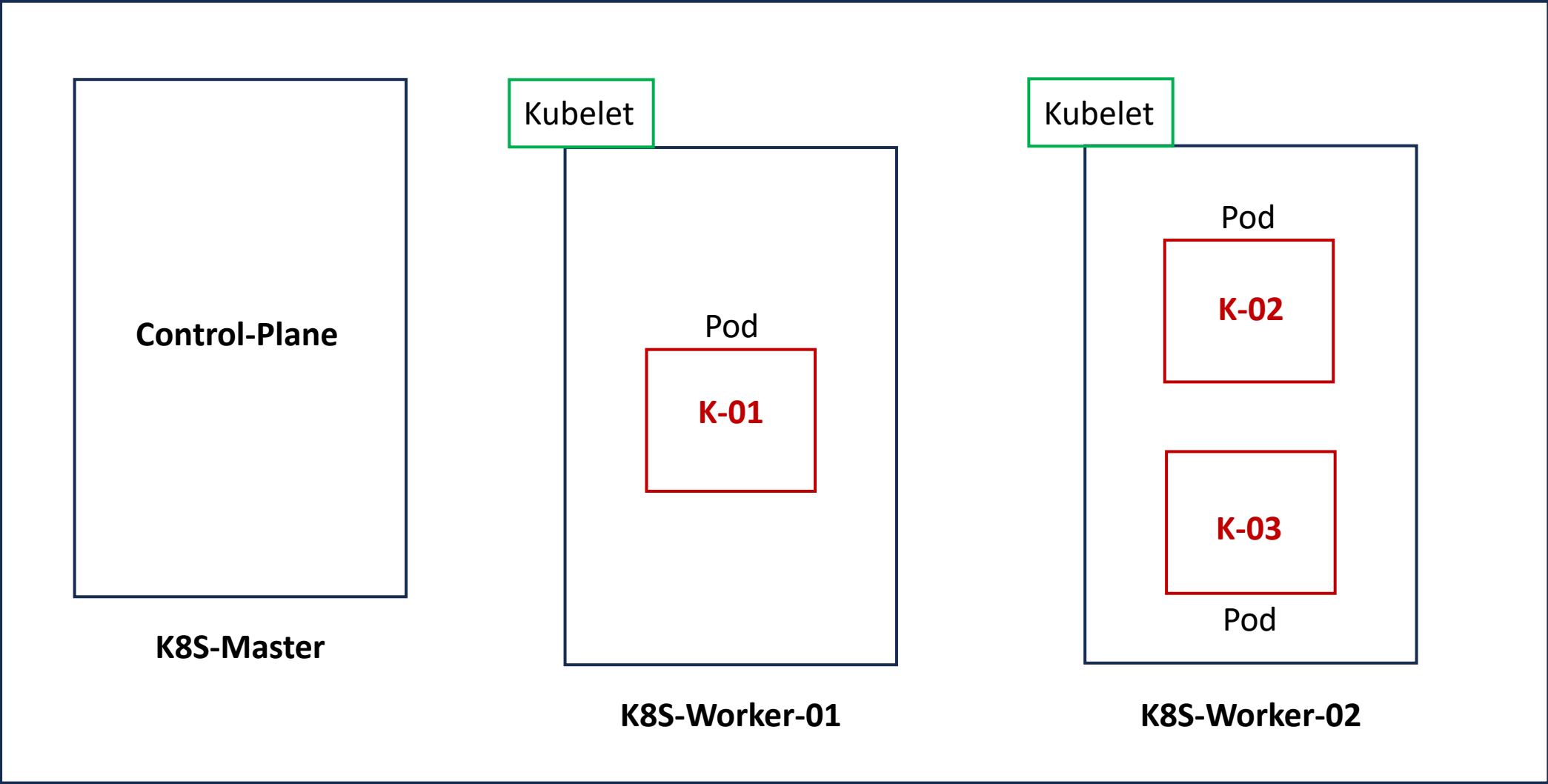


Pod

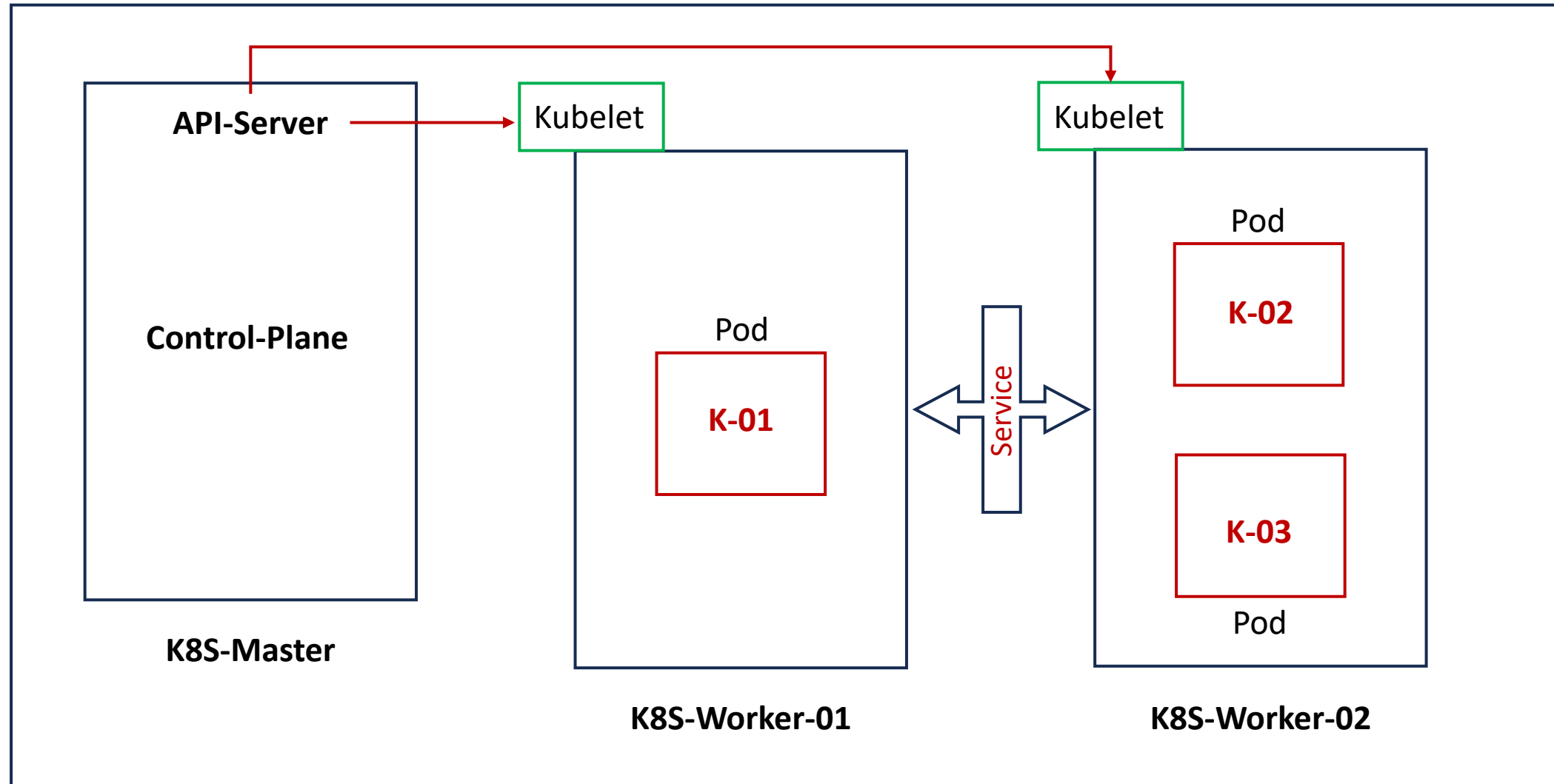
Pod

K8S-Worker-02

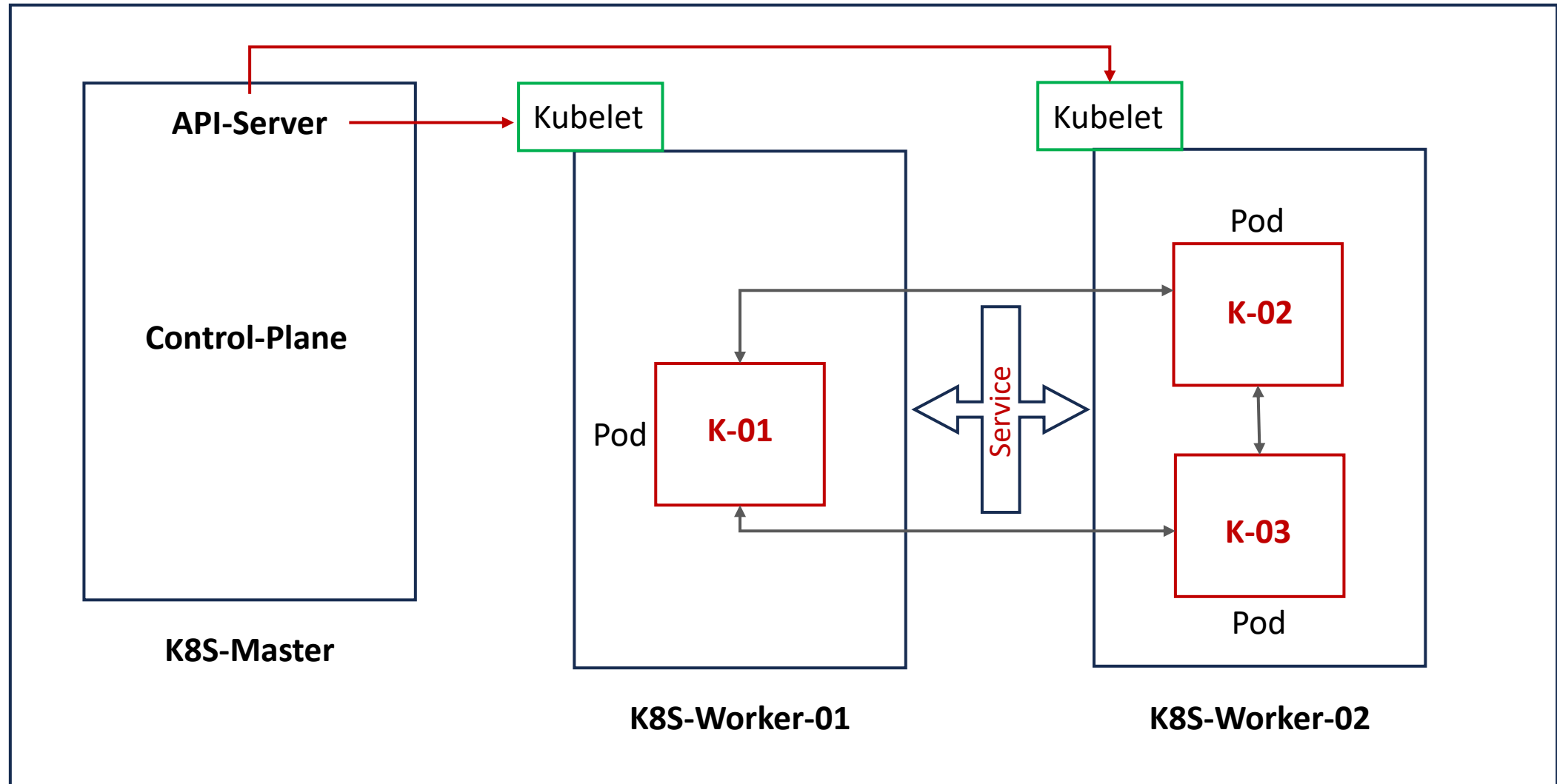
Kubernetes Cluster



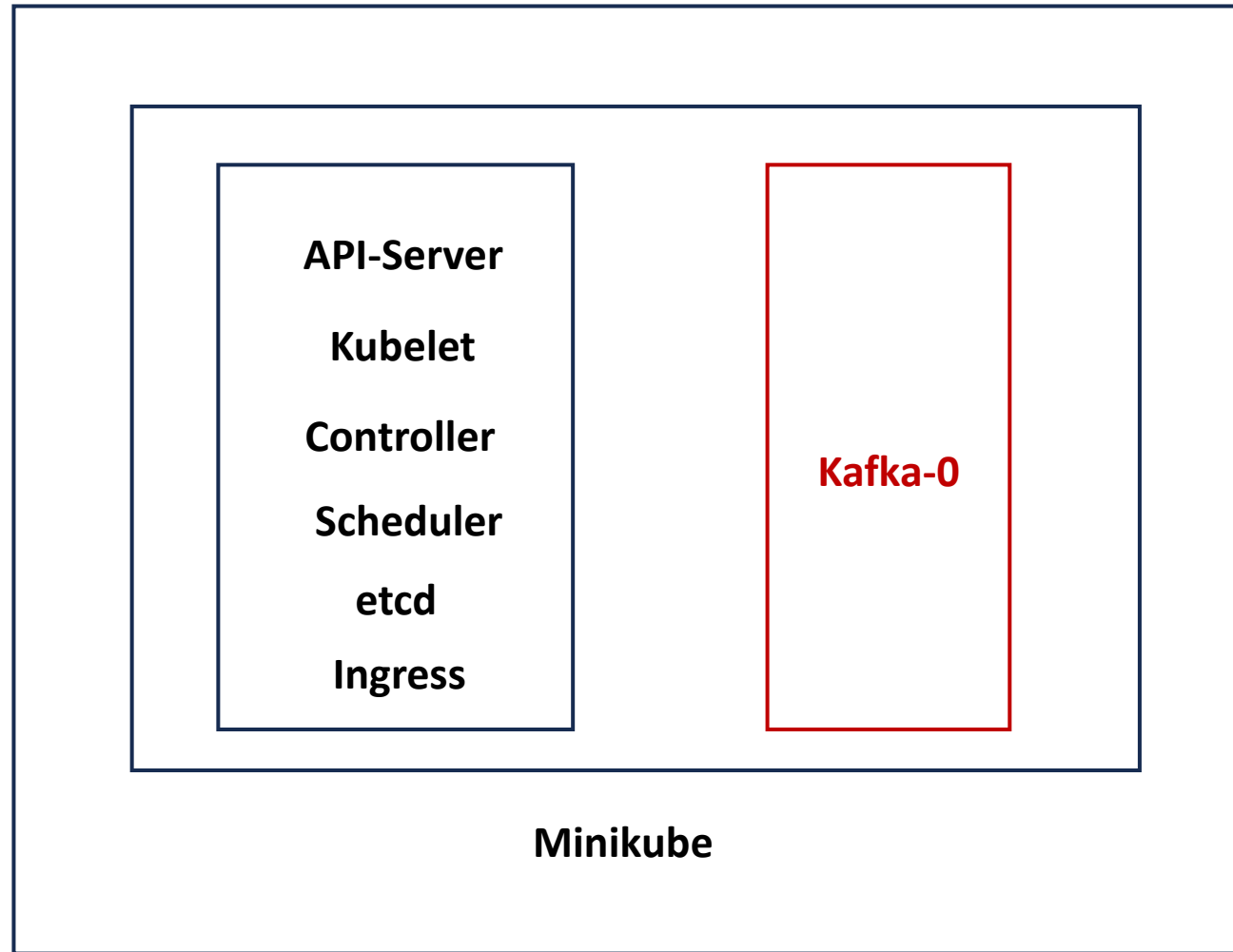
Kubernetes Cluster



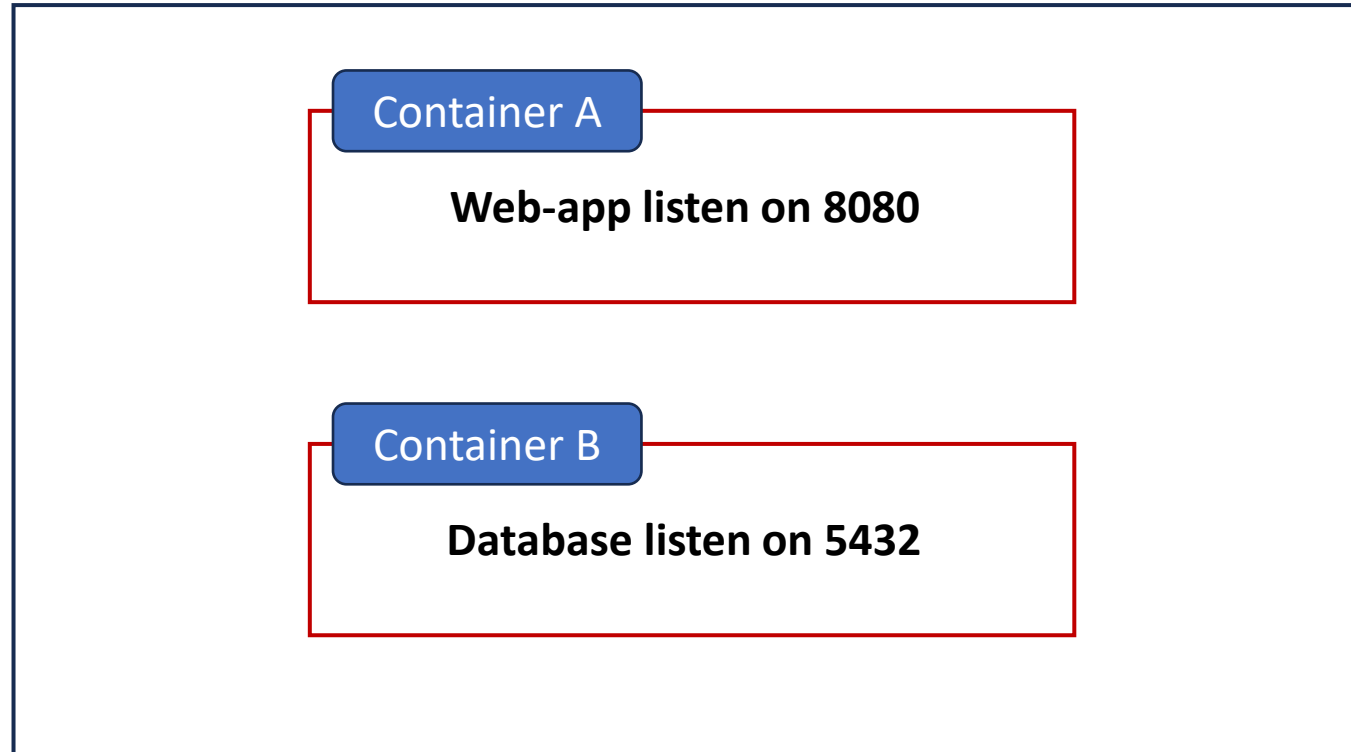
Kubernetes Cluster



Our Local System



One Pod in K8S



1. Only Pod get an IP, NOT Containers inside it
2. Container A can reach Container B at **localhost:5432**

Pod-1

Container A

Web-app listen on 8080

Pod-2

Container B

Database listen on 5432

Container A can reach Container B at

Pod-2-IP:5432

service-name:5432

Benefits of Deploying Apache Kafka on Kubernetes

Kubernetes manages:

a: Pod-level resources:

1. K8s decides which node runs each pod
2. K8s enforces CPU/memory limits
3. K8s restarts failed pods
4. K8s schedules based on available node resources

b: Infrastructure layer:

1. Network connectivity
2. Storage volumes
3. Pod lifecycle (restart policies)
4. Service discovery

Now let's go for practical jobs 🚀